Principal Component Analysis on Directional Data *

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Abstract

Directional data analysis has been emerging as an important area of statistics with its manifold applications in astronomy, biology, geology and meteorology to name a few. In this project, we mainly focus on principal component analysis (PCA) on directional data to investigate the the direction or angle which best represents the data. We discussed some basic concepts related to directional statistics which aid us in performing analysis of directional data and then introduced the principal component analysis in the context of directional data analysis. As data on direction or angle arise in specific areas beyond our study we have tried to understand the nature of principal component on directional data using simulation studies.

1 Introduction

There are various statistical problems which arise in the analysis of data when the observations are directions. We are generally more familiar working with both the magnitude and the direction of the random vector $\mathbf{x} = (x_1, x_2, \ldots x_p)'$ whereas in this review, we are concentrating only in the direction of $\mathbf{x}$. Directional data analysis has been emerging as an important area of statistics with its manifold applications in astronomy, biology, geology and meteorology to name a few. The directions are regarded as points on the circumference of a circle in two dimensions or on the surface of a sphere in three dimensions. In general, directions may be visualized as points on the surface of a hypersphere but observed directions are nothing but angular measurements. We will denote a random direction in p dimensions as $\boldsymbol{L}$, where $\boldsymbol{L}^T\boldsymbol{L} = 1$. Thus, the unit vector I takes values on the surface of a p-dimensional hypersphere S_p having unit radius and centre at the origin. It is convenient

*As a course project for MTH514A 2021, Even semester mathematically to write the density of $\boldsymbol{L}$ in terms of spherical polar coordinates $\boldsymbol{\theta}^T = (\theta_1, \theta_2, \ldots, \theta_p)$ with the help of the transformation

$$\boldsymbol{L} = u(\boldsymbol{\theta}), \quad 0 \leq \theta_i \leq \pi, \; i = 1, 2, \ldots, p-2, \; 0 \leq \theta_{p-1} \leq 2\pi$$

where

$$u_i(\boldsymbol{\theta}) = \cos(\theta_i) \prod_{j=0}^{i-1} \sin(\theta_j), \quad i = 1, \ldots, p, \quad \sin(\theta_0) = \cos(\theta_p) = 1$$

We mainly deal with two forms of directional data namely ; **axial** and **circular** or **spherical** data. We will briefly discuss them here :

1.1 Circular Data and Spherical Data

It is data that is measured on a circle in degrees or radians. It is fundamentally different from linear data due to its periodic nature ($0° = 360°$). Circular data arises in a large variety of research fields. Here the sample space is essentially a circle. The most fundamental question that can be asked of a sample of circular data is whether it suggests that the underlying population is uniformly distributed around the circle, or whether it is concentrated around at least one preferred direction. This has motivated us to perform a principal component analysis (PCA) on such data in order to gauge the "direction" or the "angle" which best represents the data. The results of the analysis on simulated circular data have been presented in the final section of this report.
In general, circular data can be extended to $p-$dimensions, where the data may be represented as points on a $p-$dimensional sphere $S_p = \{\boldsymbol{L} : \boldsymbol{L}^T\boldsymbol{L} = 1\}$ with unit radius and centre at the origin. These are typically referred as spherical data.

1.2 Axial Data

Data relating to the angular position of random lines which do not have a natural orientation associated with them, or in which neither end can be identified as the starting point, are measured in terms of angles, in radians (degrees), with a range of possible values $[0, \pi)$. We call such data axial data. Essentially axial data is just circular data observed modulo π. Assume that the axial random variable $\Theta, \Theta \in [0, \pi)$, has an absolutely continuous distribution. Any probability density function $f(\Theta)$ for an axial random variable Θ should then satisfy the following properties :

(i) $f(\theta) \geq 0$

(ii) $f(\theta) = f(\theta + \pi)$

(iii) $\int_0^\pi f(\theta)d\theta = 1$

Next in section 2, we have introduced some basic concepts in directional data analysis and later is this section we have introduced some models for spherical data and axial data which we will require in later sections. In section 3, we have discussed how PCA can be implemented in directional data analysis. In section 4, we have studied the implications and interpretations of PCA through simulated examples. Lastly in section 5, we have discussed few scenarios where PCA on directional data can be implemented in real life situations.

2 Basic Concepts

In this section, we consider some of the basic probability distributions defined on the $p-$dimensional hyper-sphere S_p. Let $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots$ be points on S_p. We first state some relevant concepts and results related to directional statistics:

(I) The location of these points can be summarized by their sample mean vector in $\mathbb{R}^p$, given by

$$\bar{\boldsymbol{x}} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i$$

Define

$$\bar{\boldsymbol{x}}_{\mathbf{0}} = \bar{R}^{-1}\bar{\boldsymbol{x}}$$

where $\bar{R} = \|\bar{\boldsymbol{x}}\|$.

The unit vector $\bar{\boldsymbol{x}}_0$ is called the *mean direction* of the sample and $\bar{R}$ is *mean resultant length.*

(II) Similarly for a unit random vector $\boldsymbol{x}$ we define the *population mean resultant length* vector ρ by

$$\rho = \left(\sum_{i=1}^{p} E(x_i)^2\right)^{\frac{1}{2}} = \Big(E(\boldsymbol{x})^T E(\boldsymbol{x})\Big)^{\frac{1}{2}}.$$

When $\rho > 0$, the *population mean vector* μ is defined by

$$\mu = \frac{E(\boldsymbol{x})}{\rho}.$$

(III) The mean direction has the *equivariance* property : Let U be an orthogonal transformation (i.e. a rotation or a reflection). Then the mean direction of $U\boldsymbol{x}_1, U\boldsymbol{x}_2, \ldots, U\boldsymbol{x}_n$ is $U\bar{\boldsymbol{x}}_0$. By contrast, the mean resultant length of $U\boldsymbol{x}_1, U\boldsymbol{x}_2, \ldots, U\boldsymbol{x}_n$ is $\bar{R}$, so that $\bar{R}$ is invariant under rotation and reflection.

(IV) The mean resultant length has the following minimisation property, which is given as follows. For $\boldsymbol{a} \in S_p$, let $S(\boldsymbol{a})$ denote the arithmetic mean of the squared Euclidean distances between x_i and $\boldsymbol{a}$. Then

$$\begin{aligned} S(\boldsymbol{a}) &= \frac{1}{n}\sum_{i=1}^{n} \|\boldsymbol{x}_i - \boldsymbol{a}\|^2 \\ &= 2(1 - \bar{\boldsymbol{x}}^T \boldsymbol{a}) \\ &= 2(1 - \bar{R}\bar{\boldsymbol{x}}_0{}^T \boldsymbol{a} \end{aligned}$$

Thus subject to the condition $\boldsymbol{a}^T\boldsymbol{a} = 1$, it follows that $S(\boldsymbol{a})$ is minimized when $a = \bar{\boldsymbol{x}}_0$ and thus,

$$\min_{\boldsymbol{a}} S(\boldsymbol{a}) = 2(1 - \bar{R})$$

The quantity $2(1 - \bar{R})$ is called the *sample spherical variance*. It is to be noted that $\bar{R} \simeq 0$ when $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ are widely dispersed and that $\bar{R} \simeq 1$ when $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ are heavily concentrated. So, $\bar{R}$ measures the extent of clustering in the mean direction.

2.1 Some models for Spherical data

In this section we introduce some distributions for directional data. Most of the distributions are defined for two dimension or circular data and then extended to higher dimensions.

2.1.1 The Uniform Distribution

The uniform distribution is the most elementary distribution defined on S_p. Here the probability of a set is proportional to its $(p-1)$ dimensional area. The uniform distribution is one of the unique distribution which is invariant under rotation and reflection. Thus it follows that for a random unit vector $\boldsymbol{L}$ *uniformly distributed* on S_p, $E(\boldsymbol{L}) = 0$, and so $\rho = 0$ and the mean direction μ is not defined.
The p.d.f. of $\boldsymbol{L}$ is given as

$$f(\boldsymbol{L}) = \frac{1}{(2\pi)^{p/2}} exp\{-\frac{1}{2}\boldsymbol{L}^T\boldsymbol{L}\}.$$

If we transform $\boldsymbol{L}$ to spherical polar coordinates $\boldsymbol{\theta}$ as described in the previous section, then the p.d.f. in terms of $\boldsymbol{\theta}$ is written as

$$f(\boldsymbol{\theta}) = \frac{\Gamma(p/2)}{2\pi^{p/2}} \prod_{i=2}^{p-1} \sin^{p-i}\theta_{i-1}.$$

2.1.2 Von Mises-Fisher Distributions

The von Mises-Fisher distributions are one of the most useful distributions defined on S_p from the point of view of statistical inference. A unit random vector $\boldsymbol{L}$ has the $(p-1)-$dimensional *von Mises-Fisher* distribution $M_p(\mu, \kappa)$ if its p.d.f with respect to the uniform distribution is written as

$$f(\boldsymbol{L}; \mu, \kappa) = \left(\frac{\kappa}{2}\right)^{p/2-1} \frac{1}{\Gamma(p/2) I_{p/2-1}(\kappa)} exp\{\kappa\mu^T\boldsymbol{L}\},$$

where $\kappa \geq 0, ||\mu|| = 1$, and I_ν denotes the modified Bessel function of the first kind and order ν, given by

$$I_\nu(\kappa) = \frac{1}{2\pi}\int_0^{2\pi} \cos\nu\theta e^{\kappa\cos\theta} d\theta.$$

2.2 Models for Axial data

2.2.1 Bingham Distributions

The Bingham distributions are used when the axial data sets do not display rotational symmemtry. The densities of these distributions are given as

$$f(\pm\boldsymbol{L}; A) = {}_1F_1\left(\frac{1}{2}, \frac{p}{2}, A\right) exp\{\boldsymbol{L}^T A \boldsymbol{L}\},$$

where the hypergemoetric function ${}_1F_1\left(\frac{1}{2}, \frac{p}{2}, .\right)$ of matrix argument is defined in (Muirhead (2009), Section 7.3).

2.2.2 Angular Central Gaussian Distribution

The normalising constant in the Bingham distributions being complicated, performing maximum likelihood estimation type methods becomes cumbersome. The angular central Gaussian distributions (ACG) makes life a bit easier in this regard. The probability density functions of $ACG(A)$ on S_p is written as

$$f(\pm\boldsymbol{L}; A) = |A|^{-1/2}(\boldsymbol{L}^T A \boldsymbol{L})^{-p/2},$$

where A is a symmetric $p \times p$ positive definite parameter matrix.

Here we have just introduced the distribution for our purpose of study and avoided any detailed properties of these distributions. For additional reading we recommend readers to K. V. Mardia and Jupp (2009).

3 Principal Component Analysis

Let $\boldsymbol{L_i}$, $i = (1)n$ be a random sample on S_p. We define $\boldsymbol{a}$ as the *first principal direction* which mzes the angles θ_i between $amiand\boldsymbol{L_i}$, $i = (1)n$ and $\boldsymbol{a}^T\boldsymbol{a} = 1$. This minimization is equivalent to maximization of the following expression:

$$Q = \sum_{i=1}^{n} \cos^2 \theta_i = \sum_{i=1}^{n} (\boldsymbol{a}^T \boldsymbol{L_i})^2 \tag{1.1}$$

Let $\mathbf{T} = \sum_{i=1}^{n} \boldsymbol{L_i} \boldsymbol{L_i}^T$ be the matrix of sums of squares and products. Suppose , $\lambda_1, ..., \lambda_p$ are the eigenvalues of $\mathbf{T}$ and $\gamma_{(1)}, .., \gamma_{(p)}$ are the corresponding eigenvectors.
Here $\lambda_1 > \lambda_2 > .. > \lambda_p$. Using the fact $\boldsymbol{L_i}^T \boldsymbol{L_i} = 1$, we get

$$tr(\mathbf{T}) = n = \lambda_1 + ... + \lambda_p \tag{1.2}$$

Now maximizing (1.1) is same as maximizing

$$Q = \boldsymbol{a}^T \mathbf{T} \boldsymbol{a}$$

subject to $\boldsymbol{a}^T \boldsymbol{a} = 1$.
Now we know that $\max Q = \lambda_1$ for $\boldsymbol{a} = \gamma_{(1)}$ which represents *first principal direction.* λ_1 measures the *concentration* around this axis.
Similarly , the *second principal direction* is $\boldsymbol{b} = \gamma_{(2)}$ which is perpendicular to $\boldsymbol{a}$. We can continue like this. For further study we refer to K. Mardia, Kent, and Bibby (1979).
If $\lambda_1 \doteq ... \doteq \lambda_p$, the angular distances will be almost the same for all $\boldsymbol{a}$. Hence , in this case , the sample distribution is approximately uniform.
The MLEs of the parameters in the Bingham distribution can be easily tied up with these eigenvalues and eigenvectors. One of the many solutions that this brings to various inferential problems can be the use of the following criterion for testing uniformity for axial data (K. Mardia et al., 1979).

$$\frac{1}{2} p(p+1) \sum_{i=1}^{n} \left(\lambda_i - \frac{n}{p} \right)^2 \sim \chi^2_{p(p+1)/2-1}$$

4 Simulation Study

Now we apply PCA on directional data simulated from different distributions. First consider the case where data are coming from a two dimensional uniform circular distribution with additional two dimension consisting minor errors. In this case we expect that first two principal component should capture maximum data variability. Here we have produced the proportion of variation captured by the principal components and all the two dimensional projection of the sample principal components. In addition we have also repeated the same experiment where data are coming from two dimensional rectangular central Gaussian function and von Mises-Fischer distribution with different parameters and additional two dimension consisting some errors. Summary of the study is given below.

Table 1: Proportion of total variability captured(cumulative) by the Principal components

Distributions	Error Distribution	1st PC	2nd PC	3rd PC	4th PC
Circular Uniform	$N_2(\mathbf{0},\ 0.05\ \mathbb{I})$	52.32	99.50	99.76	100
Circular Uniform	$N_2(\mathbf{0}, 0.5\ \mathbb{I})$	36.82	72.10	86.44	100
ACG $(A = Diag(10, 10))$	$N_2(\mathbf{0}, 0.05\ \mathbb{I})$	50.73	99.50	99.76	100
ACG $(A = Diag(100, 1))$	$N_2(\mathbf{0}, 0.05\ \mathbb{I})$	90.63	99.50	99.76	100
VMF $(\boldsymbol{\mu} = (1/\sqrt{2}, 1/\sqrt{2})$ $\kappa = 1)$	$N_2(\mathbf{0}, 0.05\ \mathbb{I})$	54.37	99.47	99.74	100
VMF $(\boldsymbol{\mu} = (1/\sqrt{2}, 1/\sqrt{2})$ $\kappa = 10)$	$N_2(\mathbf{0}, 0.05\ \mathbb{I})$	90.23	99.52	99.76	100
VMF $(\boldsymbol{\mu} = (1/2, 1/2, 1/2, 1/2, 1/2)$ $\kappa = 10)$	-	82.09	88.35	94.26	100

From the table above it is evident that first few principal components capture most of the variability in the data when the actual data has lesser dimension than its actual dimension. In case of the circular uniform distribution where error has low variability, the actual data has more variability in the first two dimension. This has been captured by the first two principal components. Similarly same results are obtained when the data are coming from two dimensional angular central Gaussian distributions with similar variances in both dimensions and von Mises-Fischer distribution with small value of concentration parameter(κ). In the case where error variability is not small compared to the actual data variability there first two principal components failed to capture most of the data variability as expected. Now when we take variance of one component of angular central Gaussian distribution to be much higher than the other component then the variability of the data is mostly in one direction. So in this case only one principal component captured more than 90% of the data variability. And taking high value of concentration parameter in von Mises-Fischer distribution will concentrate the observations, in this case also only one principal component captured 90% of the data variability. Lastly we have taken data from four dimensional von Mises-Fischer distribution with high value of concentration parameter discarding the error dimensions. In this case first principal component captured 82% data variability and first two principal component captured 88% data variability.

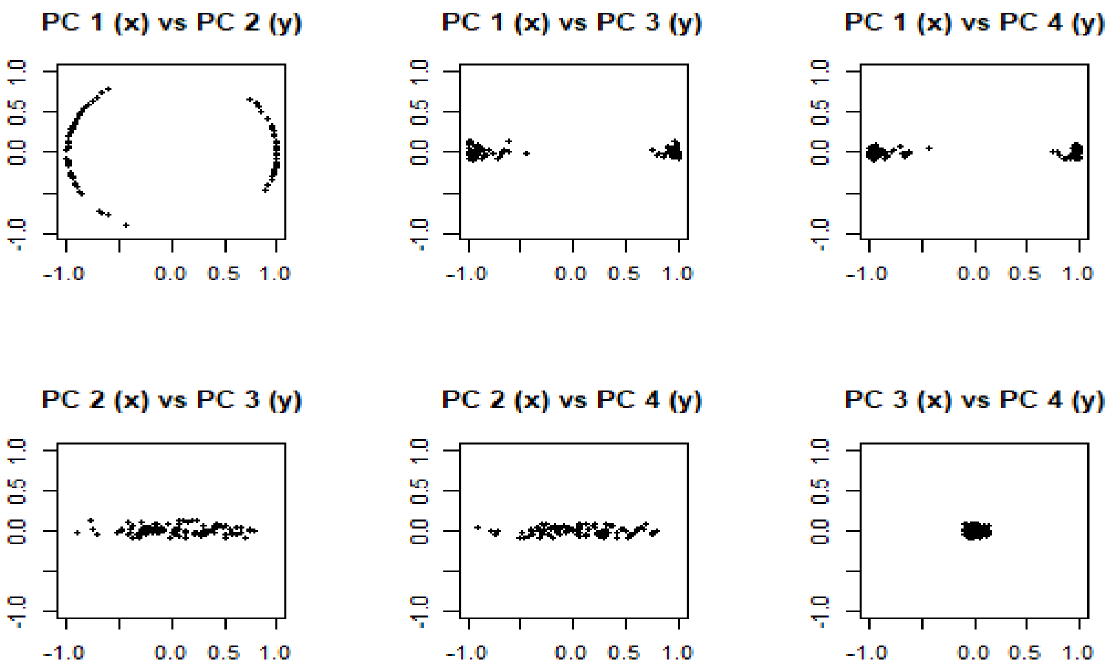

Figure 1: projection of the data

Another application of PCA is that the lower dimensional projection of the data can be used for clustering. To see this we consider the mixture of two and three von Mises-Fischer distributions with different parameters. First we consider a data from two circular von Mises-Fischer distributions with equal weights and with mean at (1,0) and at (-1,0) with concentration parameters respectively 10 and 5. In addition we have added two error dimensions. In this example it is relatively easy to identify the clusters. Here the principal components captured respectively 85.5%, 14%, 0.3% and 0.2% variability of the data. So we expect that the first PC is enough for clustering. Figure 1 shows the two dimensional projections on the principal components axes.

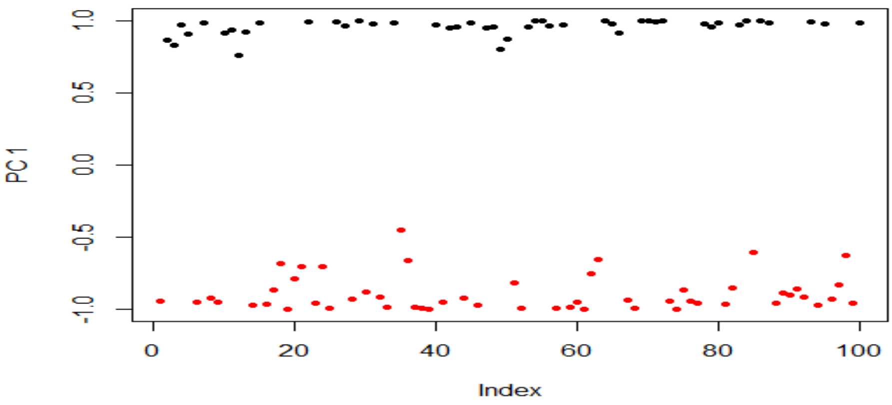

Figure 2: First sample principal components

From figure 2 it is clear that using only the first principal component we can easily identify both clusters.

Again we consider a data from two von Mises-Fischer distributions with equal weights. Now the means are at (1,0) and at (0,1) with concentration parameters respectively 20 and 10. In this case the clusters are closer to each other. Here it is not easy to identify the clusters. The principal components captured respectively 53.7%, 45.8%, 0.3% and 0.2% variability of the data. So we expect that the first PC is not enough for clustering but first two clusters are enough. Figure 3 shows the two dimensional projections on the principal components axes.

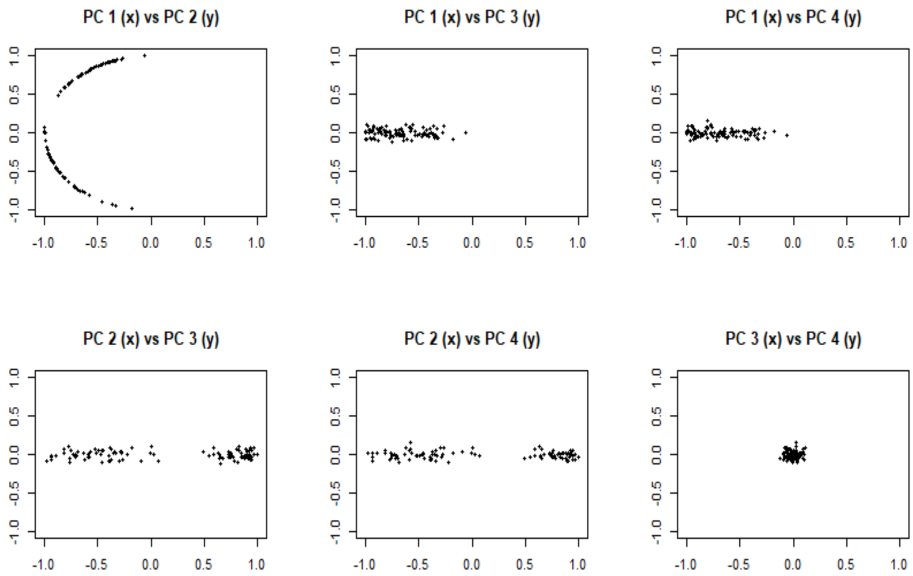

Figure 3: projection of the data

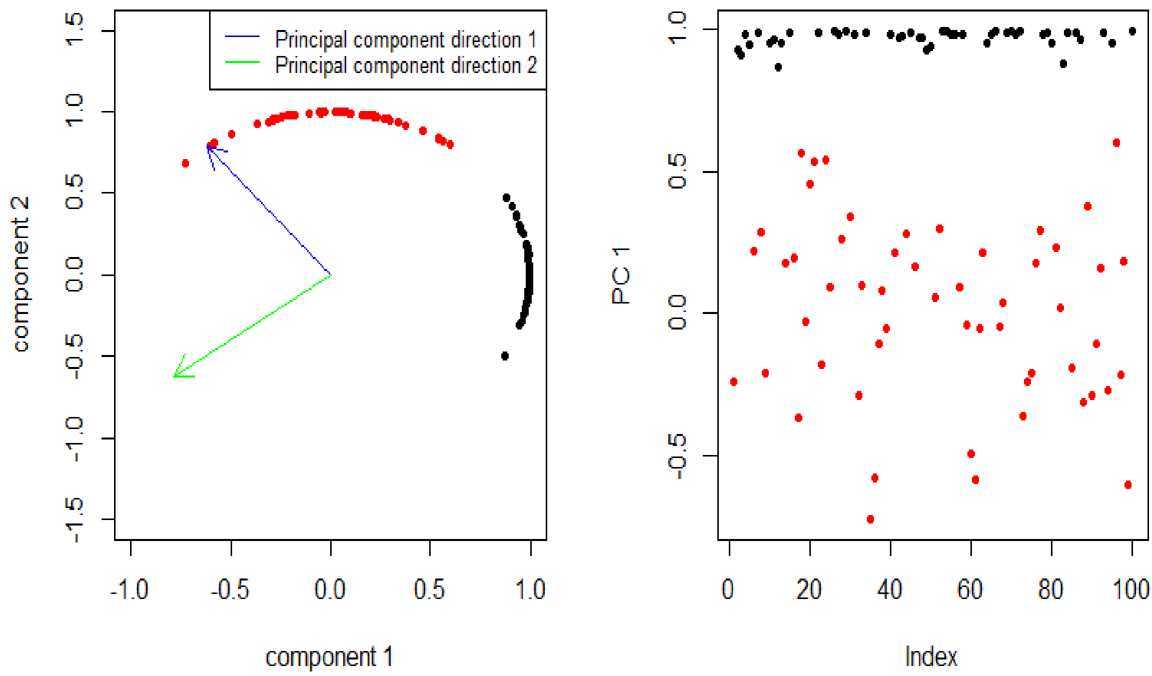

Figure 4: (a) First and second sample principal components, (a) First sample principal components

From the above figures we can say that using only one principal components the clusters are not clear. But using the first two principal components we can separate the two clusters. So we can use the two dimensional representation of the data for clustering. Note that the actual data are not marked in different colours as plotted here, so we need to determine the clusters without the colourings.

Now we consider a data from three von Mises-Fischer distributions with weights 0.3, 0.3 and 0.4. The means are at (1,0), (0,1) and at $(-1/\sqrt{2}, -1/\sqrt{2})$ with concentration parameters 20, 10 and 5 respectively for each component of the mixture. The principal components captured respectively 63.4%, 36.1%, 0.3% and 0.2% variability of the data. So we expect that the first PC is not enough for clustering. So need to take frist two principal components. Figure 5 shows the two dimensional projections on the principal components axes.

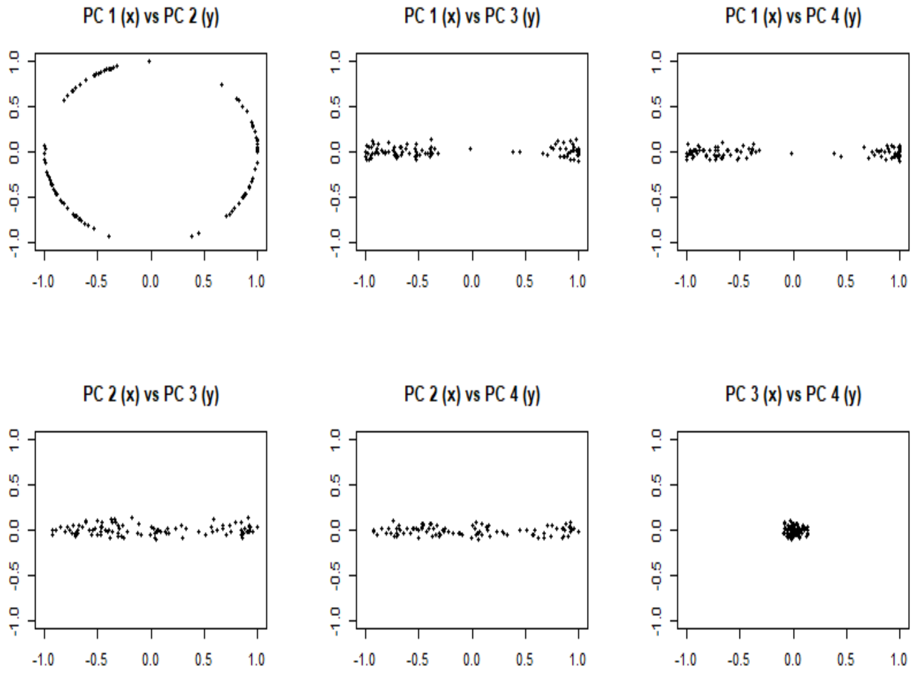

Figure 5: projection of the data

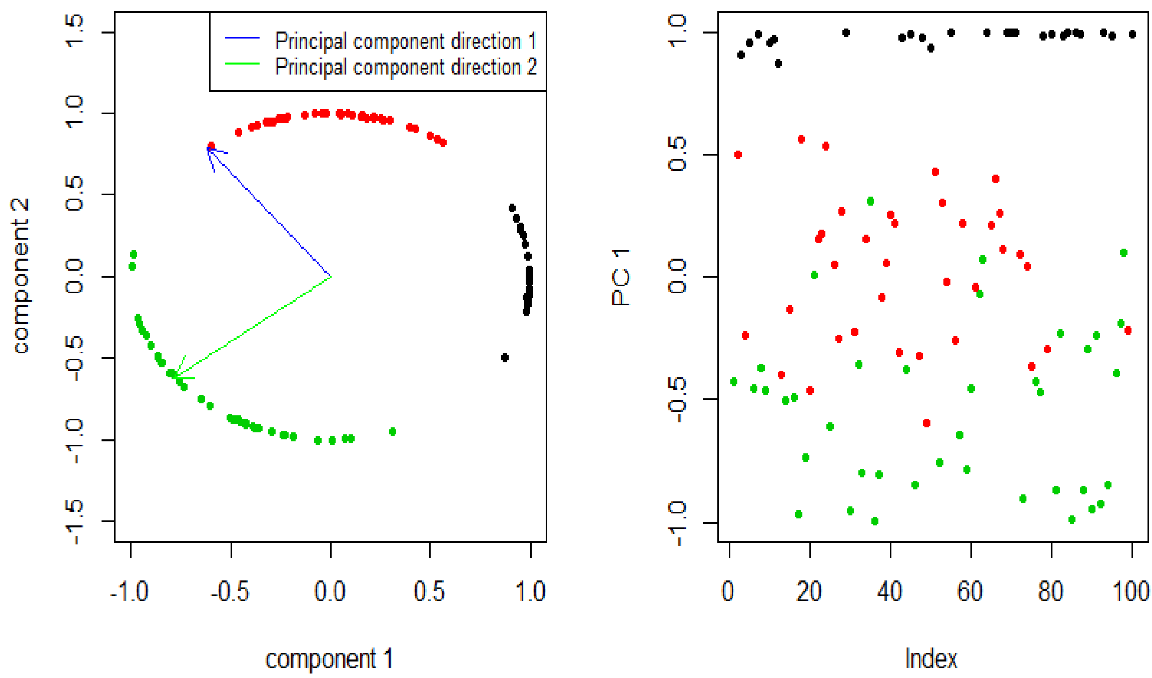

Figure 6: First sample principal components

From the above figures it is clear that using only one principal component it not possible to determine the clusters in the data. But using the first two principal components we can determine the clusters. So we can use the two dimensional representation of the data for clustering.

In all of the above examples the actual data are four dimensional but using principal component we can use one or two dimensional projection of the data to determine different clusters.

5 Application of PCA on Directional Data

The combination of PCA and the moving window approach (Davydenko and Grayver (2014)) can be applied to analyze structure of the field and suppress leveling errors in airborne geophysical data. Leveling errors typically occur during collection of geophysical data along profile lines over large areas and result in a random, possibly spatially correlated, displacement of measurements from neighboring profile lines. This creates a characteristic washboard or strip pattern perpendicular to the survey profile direction. Often geophysical data processing methods rely on the additive model i.e. measured signal is a sum of the signal we are looking for and some noise. The mean vector and covariance matrix can be used to construct flexible and efficient statistical filters. Thus employment of principal component analysis is done here.

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